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Submission #1 Report

Class: CC03

Smart E-Mobility Hub Requirements Specification

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Team Workload

This section outlines the division of work and workload distribution among team members for the Smart E-Mobility Hub requirements specification report. Each member has contributed equally to the analysis, design, and documentation of the assigned modules.

Table 0.1: Team Workload and Task Distribution

No.	Full Name	Student ID	Task	Contribution (%)
1	Võ Hoàng Phúc Duy	2452214	Hub & Identity UCs: profiles, station search, and real-time capacity/battery status. Project context, objectives, and scope.	16.67%
2	Nguyễn Khải Ca	2452159	Vehicle Rental UCs: search, reservation, pickup, and return. Stakeholder analysis, roles, and expectations.	16.67%
3	Nguyễn Phước Nhật Duy	2452205	Parking & Private EV UCs: parking reservations and charging registration. Draft and finalize the overall use-case diagram.	16.67%
4	Mai Hoàng Phương Đông	2452265	Operator Fleet & Incident UCs: dashboard, vehicle redistribution, and incident handling. Non-interactive functional requirements.	16.67%
5	Phạm Nguyễn Công Thành	2453175	Smart Charging Scheduler UCs: priority by battery level and schedule. Non-functional requirements.	16.67%
6	Nguyễn Xuân An	2452023	What-if Simulation UCs: hub-overload and crowded-metro scenarios with redistribution recommendations.	16.67%
			Total	**100%**

Note: The total contribution across all six team members equals 100%, with each member contributing an equal share of 16.67% to the project.



1 Introduction

1.1 Context

In Ho Chi Minh City, people, especially students, increasingly use Metro Line 1 for transportation and may continue their journeys using shared electric motorcycles, electric bicycles, and other forms of electric mobility. This creates a need for a management system that not only manages electric vehicles but also coordinates parking capacity, charging resources, and mobility demand across different locations.

1.2 Objective

The objective of this project is to design a software system called **Smart E-Mobility Hub** for managing and coordinating a network of electric mobility resources within the VNU-HCM Urban Area. The system aims to support users, drivers, staff, and operators in accessing and managing mobility resources efficiently.

1.3 Scope

The scope of the Smart E-Mobility Hub system includes:

- Hub and Identity Management;
- Vehicle Rental;
- Parking and Private EV Management;
- Operator Fleet and Incident Management;
- Smart Charging Scheduling; and
- What-if Simulation.

The system focuses on managing and coordinating electric mobility resources within the VNU-HCM Urban Area, including vehicles, Mobility Hubs, parking spaces, and charging resources.

The project also includes defining the system's functional and non-functional requirements, developing the system use-case model, and implementing an MVP to demonstrate the core functionality of the Smart E-Mobility Hub system.

1.4 Stakeholders

Table 1.1 summarizes the key stakeholders of the Smart E-Mobility Hub system, outlining their operational roles and expectations.

Table 1.1: System Stakeholders, Roles, and Expectations

Stakeholder	Role	Expectation
Student	End user uses Smart E-Mobility Hub network for vehicles transportation, parking and charging.	Convenient access to available vehicles and hubs, accurate availability and battery information, reliable reservation, efficient parking and charging system.
Operator	Operational user monitoring Smart E-Mobility Hub network and coordinating incidents, resources usage.	Clear network visibility, real-time status information, handling operational incidents, charging scheduling, vehicles redistribution, optimal resources allocation.

2 Use Case Model

2.1 Actors

ID	Actor	Responsibilities
ACT-01	Student	Manages a profile; finds Hubs and shared vehicles; manages rentals; reports vehicle problems; and manages private-EV parking and charging requests.
ACT-02	Operator	Monitors the Hub network; coordinates redistribution and incidents; manages charging schedules; and evaluates What-if Simulations.

Table 2.1: System Actors

2.2 Use Case Diagram

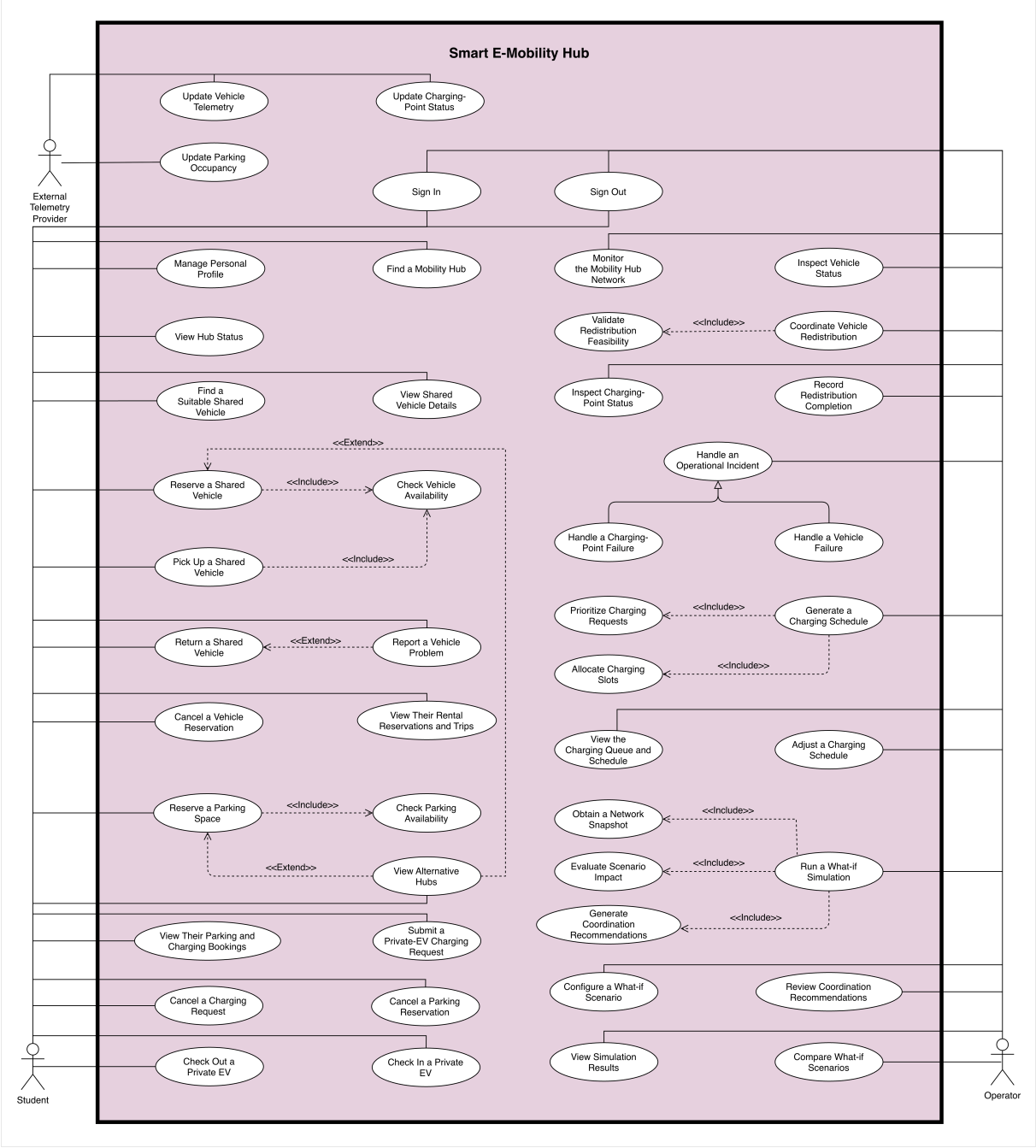


Figure 2.1: Use Case Diagram for Smart E-Mobility Hub

View the high-resolution use case diagram

2.3 Use Case Catalogue

The catalogue preserves the complete system scope. The later chapters give full scenarios only for representative workflows with substantial validation, allocation, scheduling, conflict, or failure-handling logic. A catalogue-level entry remains part of the system even when it does not have a separate detailed scenario.

Table 2.2: Complete Use Case Catalogue

ID	Use Case	Primary Actor	Purpose
UC-A01	Sign In	Student, Operator	Authenticate a user and establish a role-appropriate session.
UC-A02	Sign Out	Student, Operator	End the current session without changing operational records.
UC-H01	Manage Personal Profile	Student	View and update permitted profile information.
UC-H02	Find a Mobility Hub	Student	Search for and locate a suitable nearby Hub.
UC-H03	View Hub Status	Student	View current Hub operating status and resource capacity.
UC-R01	Find a Suitable Shared Vehicle	Student	Search for shared vehicles by preference and location.
UC-R02	View Shared Vehicle Details	Student	View the condition, availability, and details of a vehicle.
UC-R03	Reserve a Shared Vehicle	Student	Reserve an eligible vehicle without conflicting allocation.
UC-R04	Pick Up a Shared Vehicle	Student	Validate pickup and start the rental of a reserved vehicle.
UC-R05	Return a Shared Vehicle	Student	Return a rented vehicle and end its rental period.
UC-R06	View My Rental Reservations and Trips	Student	View current and past reservations and trips.
UC-R07	Cancel a Vehicle Reservation	Student	Cancel an eligible reservation and release its allocation.
UC-R08	Report a Vehicle Problem	Student	Report vehicle damage, malfunction, or another operational issue.
UC-P01	Reserve a Parking Space	Student	Reserve Hub parking for a continuous time interval.

Continued on next page

Table 2.2 – *Continued*

ID	Use Case	Primary Actor	Purpose
UC-P02	Submit a Private-EV Charging Request	Student	Request charging during an eligible private-EV parking visit.
UC-P03	View My Parking and Charging Bookings	Student	View current and past parking and charging records.
UC-P04	Cancel a Parking Reservation	Student	Cancel eligible parking and linked pending/scheduled charging.
UC-P05	Cancel a Charging Request	Student	Cancel an eligible charging request without cancelling parking.
UC-P06	Check In a Private EV	Student	Confirm arrival and begin the private-EV parking session.
UC-P07	Check Out a Private EV	Student	Confirm departure and close the parking session safely.
UC-X01	View Alternative Hubs	Student	Offer suitable Hubs when the preferred Hub cannot fulfil a request.
UC-F01	Monitor the Mobility Hub Network	Operator	Monitor network-wide Hub status, metrics, and alerts.
UC-F02	Inspect Vehicle Status	Operator	Inspect a vehicle's operational state and location.
UC-F03	Inspect Charging-Point Status	Operator	Inspect charging-point availability and operating state.
UC-F04	Coordinate Vehicle Redistribution	Operator	Create feasible assignments that rebalance vehicles between Hubs.
UC-F05	Record Redistribution Completion	Operator	Record confirmed vehicle arrivals and consistent occupancy changes.
UC-I01	Handle an Operational Incident	Operator	Provide the common incident-management behaviour for operational issues.
UC-I02	Handle a Vehicle Failure	Operator	Restrict, investigate, and resolve a failed or damaged vehicle.
UC-I03	Handle a Charging-Point Failure	Operator	Restrict, investigate, and resolve a faulty charging point.
UC-C01	View the Charging Queue and Schedule	Operator	View active requests, allocations, and schedule status.
UC-C02	Generate a Charging Schedule	Operator	Prioritize requests and generate feasible charging allocations.

Continued on next page

Table 2.2 – *Continued*

ID	Use Case	Primary Actor	Purpose
UC-C03	Adjust a Charging Schedule	Operator	Manually revise or override a generated schedule.
UC-W01	Configure a What-if Scenario	Operator	Define the conditions and period for a simulation scenario.
UC-W02	Run a What-if Simulation	Operator	Evaluate a scenario against an isolated network snapshot.
UC-W03	View Simulation Results	Operator	Review measures and visualizations from a completed run.
UC-W04	Review Coordination Recommendations	Operator	Review actions proposed from simulation outcomes.
UC-W05	Compare What-if Scenarios	Operator	Compare completed runs or a run with its baseline.
UC-S01	Check Vehicle Availability	Internal	Validate vehicle condition and conflicting commitments.
UC-S02	Check Parking Availability	Internal	Validate capacity across every slot of a requested interval.
UC-S03	Validate Redistribution Feasibility	Internal	Validate vehicle, Hub, capacity, and commitment constraints.
UC-S04	Prioritize Charging Requests	Internal	Order eligible requests using configured charging rules.
UC-S05	Allocate Charging Slots	Internal	Allocate non-conflicting charging points and intervals.
UC-S06	Obtain a Network Snapshot	Internal	Capture a consistent timestamped simulation baseline.
UC-S07	Evaluate Scenario Impact	Internal	Calculate predicted service and resource effects.
UC-S08	Generate Coordination Recommendations	Internal	Produce feasible actions or report that no intervention is required.

Detailed scenario set. The representative full specifications are UC-H01; UC-R01–UC-R08; UC-P01–UC-P07; UC-F01–UC-F02; UC-F04–UC-F05; UC-I02; UC-C01–UC-C03; and UC-W01–UC-W05. Together they cover every major module and the system's principal validation, allocation, concurrency, safety, scheduling, and simulation decisions.

2.4 Use Case Relationships

Table 2.3: Include and Generalization Relationships

Type	Source	Target	Purpose
Include	UC-R03	UC-S01	Confirm availability before reserving a vehicle.
Include	UC-R04	UC-S01	Confirm current eligibility before pickup.
Include	UC-P01	UC-S02	Confirm capacity for the whole parking interval.
Include	UC-F04	UC-S03	Confirm redistribution feasibility.
Include	UC-C02	UC-S04	Prioritize charging requests.
Include	UC-C02	UC-S05	Allocate feasible charging slots.
Include	UC-W02	UC-S06	Obtain a consistent simulation baseline.
Include	UC-W02	UC-S07	Evaluate effects relative to the baseline.
Include	UC-W02	UC-S08	Generate coordination recommendations.
Generalization	UC-I02	UC-I01	Vehicle-failure handling specializes operational incident handling.
Generalization	UC-I03	UC-I01	Charging-point failure handling specializes operational incident handling.

UC-X01 conditionally extends parking and other capacity-constrained selection flows when the preferred Hub cannot fulfil a request. Operator participation in UC-I02 and UC-I03 is inherited from UC-I01.

3 Authentication and Hub & Identity Use Cases

This chapter specifies the profile-management, Hub-search, and Hub-status workflows from the Authentication and Hub & Identity catalogue in section 2.

3.1 UC-H01 — Manage Personal Profile

Primary actor	Student (user/driver)
Goal	View and update permitted profile information without changing the Student's identity or role.
Preconditions	The Student is authenticated and has an existing profile.
Trigger	The Student selects Profile.
Priority	High
Assumption	A working internet connection is available.
Postconditions	Valid changes are stored; otherwise the last confirmed profile is preserved.

3.1.1 Main Flow

1. The Student opens Profile and selects Edit Profile.
2. The system displays the Student's profile and identifies editable fields.
3. The Student updates permitted profile information, such as contact details.
4. The Student selects Finish Editing.
5. The system validates and saves the changes.
6. The system displays "Profile updated successfully."

3.1.2 Alternative Flows

A1 — View only.
Instead of editing at step 1, the Student views the profile and exits; no data changes.

A2 — Cancel editing.
Before saving at step 5, the Student exits without saving; the system retains the existing profile information.

3.1.3 Exception Flows

E1 — Invalid information.
If a field is invalid (for example, letters in a telephone field), the system identifies it, displays the required format (for example, numbers only), and allows correction. The existing profile remains unchanged until valid data is saved.

E2 — Save failure.
The system reports the failure and does not present unconfirmed changes as saved.

3.2 UC-H02 — Find a Mobility Hub

Primary actor	Student
Goal	Find Mobility Hubs matching a location and service needs.
Preconditions	The app is open; Wi-Fi or mobile data is available.
Trigger	The Student opens the app or selects Find a Mobility Hub.
Priority	Medium
Assumption	A working internet connection is available.
Postconditions	Matching Hubs are displayed, or the Student is informed that none match.

3.2.1 Main Flow

1. The Student selects Find a Mobility Hub.

2. The system displays Hub search and filter options.
3. The Student specifies search criteria, such as location and required services.
4. The system searches for Mobility Hubs matching the criteria.
5. The system displays the matching Hubs.
6. The Student selects a Hub for further inspection.

3.2.2 Alternative Flows

A1 — Search another location.
At step 3, the Student enters a location instead of using the current GPS location. The system searches near the specified location and resumes at step 5.

3.2.3 Exception Flows

E1 — Current location unavailable.
If the system cannot obtain the Student's current location, it informs the Student and allows manual location entry.

E2 — No Mobility Hub found.
If no Hub matches the selected location and criteria, the system informs the Student, who may change the location or search range.

3.3 UC-H03 — View Hub Status

Primary actor	Student (rider)
Goal	View the latest available status and capacity information for a selected Mobility Hub.
Preconditions	The Student has opened the app and has Wi-Fi or mobile data.
Trigger	The Student selects a Mobility Hub from search results.
Priority	Medium
Assumption	A working internet connection is available.
Postconditions	Available status is displayed with a freshness timestamp; unavailable data is clearly identified.

3.3.1 Main Flow

1. The Student selects a Mobility Hub from the search results.
2. The system retrieves the selected Hub's latest status.
3. The system retrieves shared-vehicle availability.
4. The system retrieves parking occupancy and capacity.
5. The system retrieves charging-point availability.

6. The system retrieves vehicle battery information and operational status where applicable.
7. The system displays the Hub status with a data timestamp or freshness indicator.

3.3.2 Alternative Flows

A1 — Hub at full capacity.
At step 4, if no parking capacity is available, the system indicates that the Hub is full. The Student may select another Hub.

A2 — No available vehicles.
At step 3, if no vehicles are available, the system informs the Student, who may select another Hub.

A3 — Low battery availability.
At step 6, if available vehicles have low battery levels, the system indicates this so the Student can choose another vehicle or Hub.

3.3.3 Exception Flows

E1 — Battery or capacity data unavailable.
If battery or capacity data cannot be retrieved, the system reports that the information is unavailable and allows the Student to choose another Hub or continue with the remaining available information.

E2 — Unexpected data change.
If availability changes while the status is being viewed, the system refreshes the affected information and informs the Student of the update.

E3 — Vehicle or battery status unavailable.
If current status cannot be retrieved for one or more vehicles, the system marks the affected information as unavailable and informs the Student.

4 Vehicle Rental Use Cases

This chapter specifies all Vehicle Rental use cases in the catalogue in section 2. Reservation and pickup include allocation and eligibility checks; search, details, return, history, cancellation, and problem reporting are also specified below.

4.1 UC-R01 — Find a Suitable Shared Vehicle

Primary actor	Student
Goal	Find shared vehicles matching the intended journey, Hub, type, and battery requirements.
Preconditions	The system is running; the Student is authenticated and can use Vehicle Rental; vehicle and Hub status information is available.
Trigger	The Student opens Find Vehicle and enters search criteria.
Postconditions	Suitable candidates and reported availability are displayed; no vehicle is reserved.

4.1.1 Main Flow

1. The Student opens Find Vehicle.
2. The system displays available search criteria.
3. The Student selects a Hub, vehicle type, and minimum battery requirement.
4. The Student submits the search.
5. The system retrieves vehicles matching the criteria and checks reported availability and operational status.
6. The system displays suitable candidates with Hub, type, battery level, and availability.
7. The Student reviews the results.

4.1.2 Alternative and Exception Flows

A1 — No suitable vehicle at selected Hub.
The system informs the Student and allows criteria changes or another Hub; the Student may search again.

A2 — Matching vehicles become unavailable.
The system excludes unavailable or inoperable vehicles and displays any remaining candidates.

E1 — Status retrieval failure.
The system reports unavailable status and does not present unverified availability as current; the use case ends unsuccessfully.



4.2 UC-R02 — View Shared Vehicle Details

Primary actor	Student
Goal	Inspect a vehicle's identifier, type, Hub, battery, availability, and operational condition without reserving it.
Preconditions	The system is running; the Student has selected a shared vehicle; vehicle information is available.
Trigger	The Student selects a vehicle from the list.
Postconditions	The selected vehicle's details are displayed; no reservation is created.

4.2.1 Main Flow

1. The Student selects a shared vehicle.
2. The system retrieves the vehicle information and latest available status.
3. The system displays the vehicle identifier, type, current Hub, battery level, reported availability, and operational condition.
4. The Student reviews the details.

4.2.2 Alternative and Exception Flows

A1 — Vehicle becomes unavailable.
The system displays the updated availability; the Student may return to results and select another vehicle.

E1 — Details unavailable.
If the vehicle cannot be found or its current information cannot be retrieved, the system informs the Student and does not present outdated information as current; the use case ends unsuccessfully.

4.3 UC-R03 — Reserve a Shared Vehicle

Primary actor	Student
Goal	Reserve a selected shared vehicle while preventing incompatible concurrent allocation.
Preconditions	The Student is authenticated and has selected a vehicle reported as available.
Trigger	The Student requests a reservation.
Relationships	Includes UC-S01 — Check Vehicle Availability.
Postconditions	One confirmed reservation is created and the vehicle is no longer offered for incompatible commitments, or no state changes.

4.3.1 Main Flow

1. The Student selects an available vehicle and requests a reservation.
2. The system retrieves the latest vehicle state and checks availability, operational eligibility, and conflicting commitments through UC-S01.
3. The system displays the reservation summary; the Student confirms it.
4. The system atomically creates the reservation and changes the vehicle allocation state.
5. The system displays the reservation reference, status, and applicable expiry information.

4.3.2 Alternative and Exception Flows

A1 — Vehicle unavailable.
If availability changes before confirmation, the system creates no reservation and offers suitable alternatives, including vehicles at other Hubs.

A2 — Student cancels.
The request ends without changing vehicle availability.

E1 — Concurrent conflict.
If another valid commitment wins the allocation, the system rejects this request and creates no conflicting reservation.

E2 — Uncertain save.
The system does not show success until it can retrieve the original outcome; retrying must not create a duplicate reservation.

4.4 UC-R04 — Pick Up a Shared Vehicle

Primary actor	Student
Goal	Verify entitlement and vehicle condition, then start the reserved rental.
Preconditions	The Student is authenticated and has a reservation eligible for pickup.
Trigger	The Student requests pickup of the reserved vehicle.
Relationships	Includes UC-S01 — Check Vehicle Availability.
Postconditions	The reservation becomes an active rental and the vehicle becomes `In Use`, or both retain their prior confirmed states.

4.4.1 Main Flow

1. The Student opens an eligible reservation and requests pickup.
2. The system verifies ownership, reservation status, and pickup timing.
3. The system rechecks the latest vehicle availability and operational condition, accounting for the Student's reservation.

4. The system atomically starts the rental, updates the reservation, and marks the vehicle `In Use`.
5. The system confirms that the rental has started.

4.4.2 Alternative and Exception Flows

A1 — State changed.
The system uses the latest vehicle state; pickup continues only if the vehicle remains eligible.

E1 — Invalid reservation.
Pickup is denied if the reservation is expired, cancelled, ineligible, or owned by another Student.

E2 — Unsafe vehicle.
A faulted or otherwise ineligible vehicle is not unlocked and no rental starts.

E3 — Update failure.
Reservation and vehicle states are not partially advanced; the last confirmed state remains authoritative.

4.5 UC-R05 — Return a Shared Vehicle

Primary actor	Student
Goal	Validate return, end the active rental, update vehicle location and parking occupancy, and record vehicle condition.
Preconditions	The system is running; the Student has an active rental for the vehicle; Hub and parking information is available.
Trigger	The Student requests to return the rented vehicle at a Hub.
Postconditions	The rental is completed; location, parking occupancy, and condition are recorded. A faulted vehicle is not made available automatically.

4.5.1 Main Flow

1. The Student arrives at a Hub and requests to return the vehicle.
2. The system retrieves the active rental and verifies that it includes the vehicle.
3. The system retrieves and validates the return Hub and parking state.
4. The system records the vehicle's resulting condition.
5. The system ends the rental, updates the vehicle location and parking occupancy, and sets vehicle status according to its condition.
6. The system confirms the return.

4.5.2 Alternative and Exception Flows

A1 — Vehicle returned with a fault.
The system completes a valid return, marks the vehicle faulted/unavailable, and does not offer it for rental.

E1 — No matching active rental.
The system rejects the return and leaves rental status unchanged.

E2 — Return location cannot be validated.
The system informs the Student and does not complete the return; the active rental remains unchanged.

4.6 UC-R06 — View My Rental Reservations and Trips

Primary actor	Student
Goal	View personal reservations, active rental, and completed trips with their statuses.
Preconditions	The system is running and the Student is authenticated.
Trigger	The Student opens My Rentals & Trips.
Postconditions	Available rental records are displayed; no rental state is modified.

4.6.1 Main Flow

1. The Student opens My Rentals & Trips.
2. The system identifies the Student and retrieves their reservations, active rental if any, and completed trips.
3. The system displays the available records and their current statuses.
4. The Student reviews the information.

4.6.2 Alternative and Exception Flows

A1 — No rental records.
The system displays an empty-state message; the use case ends successfully.

A2 — Some record types exist.
The system displays available records and an empty-state message for categories without records.

E1 — Retrieval failure.
The system displays an error and does not present incomplete information as current; no rental data is modified.

4.7 UC-R07 — Cancel a Vehicle Reservation

Primary actor	Student
Goal	Cancel an eligible pre-pickup reservation and release its vehicle allocation.
Preconditions	The Student is authenticated; the reservation belongs to the Student, is cancellable, and has not become an active rental.
Trigger	The Student selects Cancel Reservation for a reservation.
Postconditions	The eligible reservation is cancelled and its allocation released; no active rental is ended.

4.7.1 Main Flow

1. The Student opens a reservation and selects Cancel Reservation.
2. The system verifies ownership and cancellation eligibility and displays a confirmation request.
3. The Student confirms cancellation.
4. The system cancels the reservation and releases the allocation according to the vehicle's current state.
5. The system displays confirmation.

4.7.2 Alternative and Exception Flows

A1 — Cancellation not confirmed.
The system leaves the reservation unchanged; the Student returns to its details.

E1 — Reservation ineligible.
If already cancelled, expired, or otherwise ineligible, the system displays its current status and makes no allocation change.

E2 — Rental already started.
The system rejects cancellation and directs the Student to the vehicle return process; the active rental is unchanged.

E3 — Save failure.
The system does not release the allocation and informs the Student that cancellation was unsuccessful; the previous state remains authoritative.

4.8 UC-R08 — Report a Vehicle Problem

Primary actor	Student
Secondary actor	Operator
Goal	Submit a vehicle problem linked to the vehicle and rental context for Operator review.
Preconditions	The system is running; the Student is authenticated; the vehicle can be identified and relevant rental context is available when applicable.
Trigger	The Student selects Report Vehicle Problem while using or returning a shared vehicle.
Postconditions	An incident report linked to the vehicle and relevant rental context is created for Operator review.

4.8.1 Main Flow

1. The Student selects Report Vehicle Problem.
2. The system identifies the vehicle and available rental context and displays the report form.
3. The Student enters a problem description and submits the report.
4. The system validates the required information and creates an incident linked to the vehicle and rental context.
5. The system makes the incident available for Operator review and confirms submission.

4.8.2 Alternative and Exception Flows

A1 — Report during return.
The system opens reporting with the current vehicle and rental context; after submission, the return flow records the resulting condition.

E1 — Required information missing.
The system identifies missing information; the Student supplies it and resubmits.

E2 — Vehicle or rental context invalid.
The system does not create a mislinked incident and informs the Student that submission is unavailable.

E3 — Save failure.
The system reports the failure and does not confirm a report that was not recorded.

5 Parking & Private EV Use Cases

This chapter specifies all parking and private-EV use cases in the catalogue in section 2.

5.1 Scenario Assumptions

- Parking arrival and departure selections use configured time points (15-minute spacing by default). A reservation covers the continuous half-open interval from arrival up to, but not including, departure; capacity must be available in every covered slot.
- Charging requests are linked to the same Student, vehicle, Hub, and confirmed parking reservation or active parking session. A requested charging interval is a scheduling window, not a charging allocation; it must fit within parking but need not use 15-minute increments.
- A parking reservation may be cancelled before check-in. Cancelling it also cancels linked pending or scheduled charging requests; cancelling a charging request alone leaves parking unchanged.
- Check-in and checkout are confirmed through the application; physical events may be simulated. Actual arrival and departure times are recorded without rounding. Check-in must fall within the configured check-in window; no additional grace period is assumed.
- Checkout requires active charging to end. It cancels unused linked pending or scheduled charging requests. Completed charging records remain unchanged.

5.2 UC-P01 — Reserve a Parking Space

Primary actor	Student
Goal	Reserve parking at a selected Hub for a specified continuous interval.
Preconditions	The Student is authenticated and authorized; parking policies are configured.
Trigger	The Student selects the parking-reservation function.
Relationships	Includes UC-S02 — Check Parking Availability; UC-X01 extends this use case when alternatives are requested.
Postconditions	One reservation allocates every covered slot, or no reservation/allocation is created.

5.2.1 Main Flow

1. The system displays Hubs, current parking availability, vehicle fields, and date/time selectors using configured time points (15-minute spacing by default).
2. The Student selects a Hub and date(s), enters vehicle information, and chooses arrival and departure. Offered departures must be later than arrival and feasible under reservation rules.
3. The system validates required fields and permitted time points, and requires departure to be later than arrival.

4. The system performs UC-S02 for the whole interval; every covered slot must have capacity.
5. The system displays availability for the entire interval and a Hub, vehicle, and time summary. No capacity is allocated yet.
6. The Student confirms the request.
7. The system rechecks authorization and capacity, then atomically allocates all covered slots as one reservation. Concurrent requests cannot overbook capacity; partial allocation is prohibited.
8. The system displays the reference, Hub, vehicle, interval, and `Confirmed` status.

5.2.2 Alternative Flows

A1 — Interval unavailable.
If a covered slot lacks capacity at step 4 or 7, the system rejects the whole interval without partial allocation. If requested, UC-X01 finds alternatives for the same interval. On selecting another Hub, refresh its time points and resume at step 3. The Student may instead revise the request or end without a booking.

A2 — Request changed.
Return to step 2, refresh offered times, require a new departure if the old one is no longer feasible, and repeat validation and the full interval check.

A3 — Student cancels.
Before step 7, end without creating a reservation or allocating capacity.

A4 — No times available.
Explain whether no arrival or feasible departure times are available. The Student may change Hub, date, or arrival, or end without booking.

5.2.3 Exception Flows

E1 — Invalid or missing input.
Identify invalid fields, off-grid times, departure not later than arrival, or rule violations. Return to step 2; allocate nothing.

E2 — Session expires or permission is lost.
Stop processing and require authentication or report denied access. Reload the request and recheck availability before resuming at step 3.

E3 — Availability unverifiable.
If data cannot be retrieved at steps 1, 4, or 7, report this and allow retry/cancellation. Do not offer unverified times as available or confirm a booking.

E4 — Save failure or lost response.
Do not show success until the outcome is known. For an uncertain result, retrieve

the original submission before retrying; do not duplicate a booking or partially allocate slots.

5.3 UC-P02 — Submit a Private-EV Charging Request

Primary actor	Student
Goal	Record charging needs for scheduling during a private-EV parking visit.
Preconditions	The Student is authenticated and has a parking reservation or session reference; ownership and eligibility are validated during the flow.
Trigger	The Student selects the charging-request action for a parking booking or session.
Postconditions	One linked `Pending` request exists without guaranteed charging allocation, or no request is created.

5.3.1 Main Flow

1. The system retrieves the parking record and verifies that it belongs to the Student.
2. The system displays the vehicle, Hub, parking interval, and request form.
3. The Student enters current and target battery levels and a requested charging window within the parking interval. This window need not use the parking time increment and does not allocate a charging slot.
4. The system validates inputs, Hub eligibility, and the parking record's current state.
5. The system displays a summary and explains that submission does not guarantee a slot.
6. The Student confirms submission.
7. The system rechecks parking state and atomically creates one linked `Pending` request available to the scheduler. Concurrent parking cancellation cannot leave an orphan request.
8. The system displays the request reference, vehicle, Hub, requested interval, and `Pending` status.

5.3.2 Alternative Flows

A1 — Already checked in.
Use the active parking session as context, limit the requested window to the remaining permitted parking interval, and continue at step 2.

A2 — Charging points busy.
Accept a valid request as `Pending`; do not promise an immediate slot or allocate one in this use case.

A3 — Equivalent active request.
Display the existing request and status; create no duplicate.

A4 — Change or abandon.
Return to step 3 for edits or end without creating a request.

5.3.3 Exception Flows

E1 — Invalid inputs.
Battery percentages must be 0–100, target must exceed current level, and the valid requested window must fit within parking. Identify errors and return to step 3.

E2 — Ineligible parking context.
If unavailable, owned by another user, cancelled, or otherwise ineligible, create no request; explain that an eligible booking/session is required without disclosing another user's data.

E3 — Hub does not accept charging.
Explain that charging cannot be requested there. Do not create a request or cancel parking.

E4 — Session expires.
Require authentication and reload parking state before resubmission; create no request while access is invalid.

E5 — Retrieval/save failure or lost response.
Preserve confirmed parking. Reconcile the original outcome before retrying and avoid duplicate requests.

5.4 UC-P03 — View My Parking and Charging Bookings

Primary actor	Student
Goal	View personal parking reservations, sessions, and charging requests with current statuses.
Preconditions	The Student is authenticated; existing bookings are not required.
Trigger	The Student opens the personal parking and charging bookings view.
Postconditions	Requested personal records or a valid empty state are shown; no records or allocations change.

5.4.1 Main Flow

1. The system verifies the Student's session and retrieves only their parking reservations, sessions, and charging requests.

2. The system displays vehicle, Hub, interval, current status, and any charging allocation.
3. The Student selects a record for details.
4. The system verifies access and retrieves current details and linked records.
5. The system displays details and actions permitted by current state; pending, scheduled, and active charging are distinguished.
6. The Student returns to the list or closes the view.

5.4.2 Alternative and Exception Flows

A1 — No records.
Display an empty-state message. The Student may leave or start a separate use case; viewing creates no booking.

A2 — Only some record types exist.
Display available records and empty states for absent categories.

A3 — Refresh or filter.
Retrieve matching current records belonging to the Student. A changed status updates available actions; for example, active charging is not cancellable as a pending request.

E1 — Session expires.
Require authentication before retrieving or revealing personal records.

E2 — Record unavailable or inaccessible.
Display a generic message and return to a refreshed list without disclosing another Student's data.

E3 — Retrieval failure.
Display an error and offer retry; do not report failure as "no bookings." Mark any stale displayed data as not refreshed.

5.5 UC-P04 — Cancel a Parking Reservation

Primary actor	Student
Goal	Cancel an eligible parking reservation and its dependent pending or scheduled charging requests.
Preconditions	The Student is authenticated and has a reservation reference; eligibility is checked during the flow.
Trigger	The Student selects Cancel for a parking reservation.
Postconditions	Reservation and eligible linked requests are cancelled and allocations released once, or no partial cancellation occurs.

5.5.1 Main Flow

1. The system retrieves the reservation and verifies ownership.
2. The system verifies it is `Confirmed`, not checked in, and cancellable.
3. The system retrieves linked pending/scheduled requests and displays all cancellation consequences.
4. The Student confirms.
5. The system rechecks state and atomically cancels the reservation and eligible linked requests, releasing parking capacity across covered slots and dependent charging allocations without partial updates or double release.
6. The system displays the cancelled reservation and linked request statuses.

5.5.2 Alternative and Exception Flows

A1 — No linked requests.
Show only the parking cancellation; release parking allocation only.

A2 — Student declines.
End without changing records or allocations.

A3 — Already cancelled.
Display the existing result; do not release resources again.

A4 — Linked states changed.
Refresh the summary and obtain confirmation of updated consequences if requests remain cancellable.

E1 — Record unavailable or not owned.
Reject without revealing protected details; return to the Student's list.

E2 — Cancellation no longer permitted.
Display current state. A checked-in session must use UC-P07; completed or expired reservations cannot be cancelled.

E3 — Linked request cannot be cancelled.
Stop the combined operation; do not cancel parking while leaving an unresolved active linked charging session.

E4 — Session expires.
Require authentication and reload reservation/requests before reconfirmation.

E5 — Save failure or uncertain result.
Do not leave parking cancelled while eligible requests remain allocated. Reconcile before retrying; never release capacity twice.



5.6 UC-P05 — Cancel a Charging Request

Primary actor	Student
Goal	Cancel a pending or scheduled charging request while preserving its parking booking/session.
Preconditions	The Student is authenticated and has a charging-request reference; eligibility is checked during the flow.
Trigger	The Student selects Cancel for a charging request.
Postconditions	Request is cancelled and future allocation released once; parking is unchanged, or no partial change occurs.

5.6.1 Main Flow

1. The system retrieves the request and verifies ownership.
2. The system verifies that status is `Pending` or `Scheduled`.
3. The system displays any allocation to be released and states that parking remains unchanged.
4. The Student confirms.
5. The system rechecks state, marks the request `Cancelled`, and releases future charging allocation once.
6. The system displays the cancelled request and unchanged parking state.

5.6.2 Alternative and Exception Flows

A1 — Pending with no allocation.
Mark cancelled without changing charging-point capacity.

A2 — Student declines.
End without changes.

A3 — Already cancelled.
Return the existing result without releasing allocation again.

A4 — Allocation changed.
Refresh the summary and reconfirm while the request remains cancellable.

E1 — Request inaccessible.
Reject without disclosing another user's information.

E2 — Charging active or request completed/interrupted.
Reject cancellation and display current state; an active session must use the charging-interruption procedure.

E3 — Session expires.
Require authentication and reload state before confirmation.

E4 — Save failure or lost response.
Avoid partial changes; reconcile uncertain outcomes. Parking remains unaffected.

5.7 UC-P06 — Check In a Private EV

Primary actor	Student
Goal	Record arrival and start the reserved private-EV parking session.
Preconditions	The Student is authenticated, has arrived, and has a parking reservation; current validity is checked during execution.
Trigger	The Student selects Check In for the reservation.
Postconditions	One active session is linked to the reservation; actual arrival is recorded without rounding and occupancy is updated once.

5.7.1 Main Flow

1. The system retrieves the reservation and verifies access, then displays Hub, vehicle, and interval.
2. The Student confirms arrival and vehicle/Hub details.
3. The system verifies ownership, vehicle and Hub match, `Confirmed` state, and that current time is within the configured check-in window.
4. The system verifies the reserved allocation can be honoured and no session already exists.
5. The system starts one `Active` session, records actual arrival time without rounding, marks the reservation `Checked In`, and updates occupancy once while retaining commitments for the remaining reserved interval.
6. The system displays the active session and any linked charging status. Check-in does not mean charging has started.

5.7.2 Alternative and Exception Flows

A1 — Already checked in.
Display the existing session without creating another or incrementing occupancy again.

A2 — Too early.
Display when check-in becomes available; leave reservation unchanged and end the attempt.

A3 — Student declines.
End without changing reservation or occupancy.

E1 — Invalid reservation.
If unavailable, inaccessible, cancelled, expired, completed, or otherwise invalid, reject without starting a session.

E2 — Vehicle or Hub mismatch.
Explain the mismatch and return to details; do not use another reservation.

E3 — Allocation cannot be honoured.
Do not check in or add occupancy; preserve the reservation and report the operational issue.

E4 — Session expires or state changes.
Require authentication and reload. Continue only if check-in is still permitted.

E5 — Save failure or lost acknowledgement.
Do not leave a checked-in reservation without its session or count occupancy twice. Reconcile before retrying.

5.8 UC-P07 — Check Out a Private EV

Primary actor	Student
Goal	Close the parking session after resolving charging and confirming departure.
Preconditions	The Student is authenticated and has a parking-session reference.
Trigger	The Student selects Check Out before leaving the Hub.
Postconditions	Session is closed, actual departure recorded without rounding, and unused linked charging requests cancelled; allocations are released once.

5.8.1 Main Flow

1. The system retrieves the session and verifies Student and vehicle ownership.
2. The system retrieves linked requests and current charging-session state; checkout proceeds only when charging is not active.
3. The system displays a summary, including pending/scheduled requests that will be cancelled.
4. The Student confirms departure through the application.
5. The system rechecks parking/charging state, closes the session, marks the reservation `Completed`, and records actual departure time without rounding.
6. The system cancels unused pending/scheduled requests and releases occupied parking, any remaining held capacity for the visit, and charging allocations once as a consistent update. Completed charging history is unchanged.
7. The system displays the completed visit and linked request statuses.

5.8.2 Alternative and Exception Flows

A1 — Charging is active.
Ask whether the Student wants to stop charging and continue. If yes, request stop and wait for confirmation; record charging as `Completed` if target met, otherwise `Interrupted`, then show the checkout summary and require departure confirmation again. If declined, end with parking active.

A2 — Student cancels checkout.
Leave parking active. Any charging already stopped remains stopped.

A3 — Already checked out.
Display the existing result without releasing capacity again.

A4 — No linked charging requests.
Proceed with parking-only checkout.

E1 — Session unavailable or mismatched.
Reject without exposing protected information or changing occupancy.

E2 — State cannot be verified.
Do not confirm checkout or release capacity; report failure and allow retry.

E3 — Charging stop fails or is unconfirmed.
Keep parking active and do not treat the charging point as free; report for assistance.

E4 — Session expires.
Require authentication and reload both states. A previously confirmed charging stop remains effective.

E5 — Finalization fails or response is lost.
Avoid partial closure or double release. Reconcile before retrying; do not represent stopped charging as restarted.

6 Fleet Monitoring & Incident Handling Use Cases

This chapter specifies the Operator's fleet and vehicle-incident scenarios from the catalogue in section 2: network monitoring, vehicle inspection, redistribution, and vehicle-failure handling. **Scope note:** UC-I01 — Handle an Operational Incident is an abstract parent; UC-I02 specifies its vehicle-failure specialization. Charging-point monitoring (UC-F03) and charging-point failure handling (UC-I03) remain in the system catalogue but are outside this chapter's scope.

6.1 UC-F01 — Monitor the Mobility Hub Network

Primary actor	Operator
Goal	View a network-wide operational overview of vehicle distribution, parking use, charging availability, incidents, shortages, and data freshness.
Preconditions	The system is running; the Operator is authenticated and authorized; Hub and resource-status information is available.
Trigger	The Operator opens the Operator Dashboard.
Postconditions	Available network information is displayed; no vehicle, Hub, reservation, or incident state changes.

6.1.1 Main Flow

1. The Operator opens the dashboard.
2. The system verifies access permission.
3. The system retrieves latest available Hub status, vehicle distribution and availability, parking utilization, charging-point availability, and open incidents.
4. The system identifies Hubs approaching capacity limits or experiencing vehicle/resource shortages.
5. The system displays the network overview and data-freshness time for each Hub.
6. The Operator reviews the information.

6.1.2 Alternative and Exception Flows

A1 — No open incidents.
The system indicates that no operational incident is open and continues displaying the remaining network information.

A2 — Hub nearing a limit.
The system highlights the affected Hub and condition. The Operator may inspect a vehicle or begin redistribution; this use case does not change live state.

E1 — Status retrieval failure.
The system records the failure, marks affected information unavailable or shows the last known value with its freshness time, and does not present it as current.

6.2 UC-F02 — Inspect Vehicle Status

Primary actor	Operator
Goal	Inspect a vehicle's location, battery, availability, rental state, condition, and reported faults before an operational decision.
Preconditions	The system is running; the Operator is authenticated and authorized; the selected vehicle exists.
Trigger	The Operator selects a vehicle from the dashboard or fleet list.
Postconditions	Available vehicle information is displayed; no vehicle state, assignment, reservation, or rental is modified.

6.2.1 Main Flow

1. The Operator selects a vehicle.
2. The system retrieves its identifier and latest available status.
3. The system retrieves current Hub/location, battery, availability, rental state, operational condition, and reported faults.
4. The system displays the vehicle details.
5. The Operator reviews the status.

6.2.2 Alternative and Exception Flows

A1 — Reported faults.
The system displays reported faults and incident status. The Operator may begin UC-I02; vehicle details remain available.

E1 — Vehicle or status unavailable.
If the vehicle cannot be found or required current information cannot be retrieved, the system reports unavailable status and does not present missing data as current.

6.3 UC-F04 — Coordinate Vehicle Redistribution

Primary actor	Operator
Goal	Create a feasible assignment to move eligible vehicles between Hubs; creation does not assert physical arrival.
Preconditions	The Operator is authorized and current vehicle and Hub data are available.
Trigger	The Operator opens redistribution coordination.
Relationships	Includes UC-S03 — Validate Redistribution Feasibility.
Postconditions	A feasible assignment is recorded; vehicle location and Hub occupancy change only after confirmed arrival through UC-F05.

6.3.1 Main Flow

1. The system displays eligible vehicles and source/destination Hub information.
2. The Operator selects one or more vehicles and the source and destination Hubs.
3. The Operator requests validation; through UC-S03 the system checks vehicle eligibility, conflicting commitments, Hub suitability, and destination capacity.
4. The system displays feasibility and transfer details. The Operator confirms a feasible request.
5. The system creates and records a redistribution assignment without changing physical vehicle locations or Hub occupancy.
6. The system displays the assignment confirmation and state.

6.3.2 Alternative and Exception Flows

A1 — Some vehicles ineligible.
The system identifies unavailable, faulted, reserved, rented, or already committed vehicles. The Operator may remove them or choose replacements and continue with eligible vehicles.

A2 — Insufficient destination capacity.
The system rejects the proposed transfer and explains the capacity issue. The Operator may select another Hub or reduce the transfer.

A3 — Operator cancels.
Before confirmation, end without an assignment or operational-state change.

E1 — Stale or unavailable data.
Do not confirm feasibility until required current state can be validated.

E2 — Assignment save failure.
Do not confirm the assignment; vehicle locations and Hub occupancy remain unchanged.

6.4 UC-F05 — Record Redistribution Completion

Primary actor	Operator
Goal	Confirm actual vehicle arrivals for an assignment and consistently update location, assignment state, and Hub occupancy.
Preconditions	The Operator is authorized; a valid incomplete assignment exists; arrived vehicles can be identified.
Trigger	The Operator selects an assignment and confirms vehicle arrival.
Postconditions	Only confirmed movements are recorded, with location, assignment, and occupancy updated consistently.

6.4.1 Main Flow

1. The Operator opens an incomplete assignment.
2. The system displays assigned vehicles, source and destination Hubs, and assignment state.
3. The Operator confirms that all assigned vehicles arrived.
4. The system validates the vehicles against the assignment.
5. The system updates vehicle locations, marks the assignment `Completed`, and updates source/destination occupancy consistently.
6. The system displays the completion result.

6.4.2 Alternative and Exception Flows

A1 — Partial completion.
The Operator confirms only arrived vehicles. The system validates and updates location/occupancy for that subset, keeps the assignment partial or in progress, and leaves unconfirmed vehicles unchanged.

A2 — Mismatched vehicle.
The system rejects a vehicle not assigned to this transfer, identifies the mismatch, and allows correction.

E1 — Assignment unavailable or complete.
The system displays the current state and changes neither vehicle locations nor occupancy.

E2 — Completion update failure.
Do not confirm completion; record the error and preserve the last confirmed operational state.

6.5 UC-I02 — Handle a Vehicle Failure

Primary actor	Operator
Incident sources	Student report or external telemetry may provide secondary information.
Goal	Investigate a vehicle failure, protect operations, record corrective action, and restore eligibility only after resolution.
Preconditions	The Operator is authorized; an identifiable vehicle-failure incident exists; the affected vehicle can be identified.
Trigger	The Operator selects the incident.
Relationships	Specializes UC-I01 — Handle an Operational Incident.
Postconditions	Incident and eligibility reflect the last confirmed handling state; unresolved vehicles remain restricted.

6.5.1 Main Flow

1. The system retrieves the incident and displays the vehicle's location, condition, fault, evidence, rental state, and related reservations.
2. The Operator investigates the failure and requests restriction of unsafe/ineligible use.
3. The system prevents new vehicle allocations and identifies affected commitments.
4. The Operator records responses for affected users/rentals and corrective, recovery, or repair actions.
5. The system updates the incident while keeping the vehicle restricted until recovery or repair is confirmed.
6. When resolved, the Operator requests restoration of eligibility.
7. The system records incident and vehicle-state transitions and restores eligibility only after completed recovery/repair is recorded.
8. The system displays the updated incident status.

6.5.2 Alternative and Exception Flows

A1 — Active rental.
The system displays the active rental. The Operator records the required operational response before resolution; the vehicle receives no new commitments.

A2 — Pending reservations.
The system displays affected reservations. The Operator handles commitments under applicable policy, while the vehicle remains unavailable for new ones.

A3 — Repair incomplete.
The Operator records current repair/recovery status. The incident remains open or unresolved and the vehicle remains restricted.

E1 — Incident or vehicle information unavailable.
Report the error and make no eligibility or resolution change.

E2 — Incident update failure.
Do not confirm resolution or restoration; retain the last confirmed incident and eligibility states.

6.6 Related Use-Case Relationships

- UC-F04 includes UC-S03 — Validate Redistribution Feasibility.
- UC-I02 specializes UC-I01 — Handle an Operational Incident.
- A vehicle problem report (UC-R08) or telemetry event may provide information leading to a vehicle-failure incident handled by UC-I02.

7 Smart Charging Scheduling Use Cases

This chapter specifies the Smart Charging Scheduler use cases in the catalogue in section 2. Pending requests use First-Come, First-Served (FCFS) order by submission timestamp; Request ID is the deterministic tie-breaker for identical timestamps. Operator adjustments are identified as overrides to the default queue order.

7.1 UC-C01 — View the Charging Queue and Schedule

Primary actor	Operator
Supporting actor	Student (charging-request source)
Goal	View pending requests in FCFS order, current charging-point allocations, planned intervals, and unscheduled requests with reasons.
Preconditions	The system is running; the Operator is authorized; charging requests and charging-point information exist.
Trigger	The Operator opens Charging Queue & Schedule.
Postconditions	Current queue, schedule, allocations, and unscheduled requests are displayed; scheduling data is unchanged.

7.1.1 Main Flow

1. The Operator opens Charging Queue & Schedule.
2. The system verifies access permission.
3. The system retrieves active/pending requests, their submission timestamps, charging-point status, and the current schedule.
4. The system orders pending requests by submission timestamp in FCFS order; identical timestamps are ordered by Request ID. Any manual queue overrides are identified.
5. The system displays request statuses and timestamps, the schedule with assigned points and planned intervals, and unscheduled requests with reasons.
6. The Operator reviews the information.

7.1.2 Alternative and Exception Flows

A1 — No pending requests.
Display an empty-queue message and continue to show existing schedule and charging-point status.

A2 — Unscheduled requests.
Display each unscheduled request and its reason, such as no available charging point or insufficient charging time.

E1 — Scheduling-data retrieval failure.
Record and report the error; do not present incomplete data as the current schedule.

E2 — Charging-point status unavailable.
Mark point information temporarily unavailable. Do not modify the existing schedule.

7.2 UC-C02 — Generate a Charging Schedule

Primary actor	Operator
Supporting system	Smart Charging Scheduler
Goal	Generate or refresh a feasible schedule using FCFS request order, charging-duration needs, and available resources.
Preconditions	The Operator is authorized; active requests, required scheduling data, and charging-point status are available.
Trigger	The Operator selects Generate Charging Schedule or Refresh Schedule.
Relationships	Includes UC-S04 — Prioritize Charging Requests and UC-S05 — Allocate Charging Slots.
Postconditions	A valid non-overlapping schedule is stored; unscheduled requests have reasons, or the previous valid schedule remains active.

7.2.1 Main Flow

1. The Operator requests schedule generation.
2. The system verifies permission and retrieves active requests, submission timestamps, required charging durations, vehicle charging needs, point status, and capacity.
3. The system validates the scheduling data.
4. Through UC-S04, the system orders pending requests by submission timestamp (FCFS), using Request ID to break identical timestamps.
5. Through UC-S05, the system evaluates available points and intervals and assigns each feasible request in order, without conflicts and within its requested window.
6. The system marks requests without a valid allocation as `Unscheduled` with a reason.
7. The system validates and atomically saves the complete schedule.
8. The system displays allocations and unscheduled requests; the Operator reviews the result.

7.2.2 Alternative Flows

A1 — Charging unnecessary.
If the vehicle has already reached its target, mark the request `No Charging Required`, exclude it from allocation, and continue.

A2 — Insufficient charging time.
If the available time before required vehicle use is insufficient, leave the request unallocated, mark it `Unscheduled` with reason Insufficient Charging Time, and continue.

A3 — No available charging point.
If no suitable point is available, mark the request `Unscheduled` with reason No Available Charging Point and continue.

A4 — Identical submission timestamps.
Process the request with the earlier Request ID first.

7.2.3 Exception Flows

E1 — Point or vehicle data unavailable.
Stop generation, report the unavailable information, and preserve the previous valid schedule.

E2 — Scheduling timeout or calculation failure.
Terminate the operation and do not publish an incomplete schedule; the previous schedule remains active.

E3 — Schedule save failure.
Do not activate or display the generated result as current. Record and report the failure; preserve the previous valid schedule.

7.3 UC-C03 — Adjust a Charging Schedule

Primary actor	Operator
Goal	Manually adjust a request's queue position, interval, or charging-point assignment while preserving schedule validity.
Preconditions	The Operator is authorized; a current schedule and point-status information exist; the request is eligible for adjustment.
Trigger	The Operator selects a request and chooses Adjust Schedule.
Postconditions	An adjustment is stored only after validation; otherwise the previous valid schedule remains active.

7.3.1 Main Flow

1. The Operator selects a request from the current schedule.
2. The system retrieves and displays its queue position, point, and planned interval.
3. The Operator changes one or more permitted scheduling attributes and selects Save Adjustment.
4. The system validates the proposed queue order, interval, point status, conflicts, and scheduling constraints.
5. If valid, the system saves the adjustment and updates the schedule.
6. The system confirms and displays the adjusted schedule, identifying any manual queue override.

7.3.2 Alternative Flows

A1 — Queue position adjusted.
Validate the new sequence and mark it as an Operator override to the default FCFS order. Existing allocations remain unchanged unless revalidated as part of the adjustment.

A2 — Charging interval changed.
Validate the new interval against the request window and all other allocations. If conflict-free, continue to step 5.

A3 — Charging point changed.
Retrieve the selected point's latest status and verify operational availability throughout the interval. If valid, continue to step 5.

A4 — Proposed change conflicts.
Reject the adjustment, identify the conflict, and allow another interval or point. The original schedule remains unchanged.

A5 — Selected point is faulty or unavailable.
Reject the point assignment, display its status, and allow another point; preserve the original schedule.

7.3.3 Exception Flows

E1 — Point status unavailable.
The system cannot safely validate the change; report the issue and preserve the current schedule.

E2 — Database update failure.
Roll back the attempted modification, report the error, and keep the previous valid schedule active.

E3 — Concurrent schedule modification.
Detect that the edited schedule is stale, reject the outdated update, and ask the Operator to reload the latest version.

8 What-if Simulation Use Cases

This chapter specifies all What-if Simulation use cases in the catalogue in section 2: configuration, isolated execution, results, recommendations, and comparison.

8.1 Simulation Scope and Assumptions

- A simulation runs against a consistent, timestamped copy of network state. Its hypothetical events and state transitions affect only the isolated simulation state.
- Simulation must not modify live vehicles, reservations, parking occupancy, charging sessions or schedules, incidents, or redistribution assignments. Reviewing a recommendation also does not apply it to live operations.
- Results retain the run identifier, scenario assumptions, baseline timestamp, data freshness, and simulation period so that predicted values can be interpreted against their source state.
- Supported hypothetical conditions include increased Metro arrivals, full or offline Hubs, increased charging demand, charging-point failures, and excessive vehicle concentration.

8.2 UC-W01 — Configure a What-if Scenario

Primary actor	Operator
Goal	Define and validate hypothetical operating conditions, resources, assumptions, and simulation period.
Preconditions	The Operator is authorized; Hubs/resources are registered; simulation configuration rules are available.
Trigger	The Operator selects Create What-if Scenario.
Postconditions	A valid configuration is saved as `Ready to Run`; no simulation has run and no live state has changed.

8.2.1 Main Flow

1. The Operator opens the simulation workspace and selects Create What-if Scenario.
2. The system verifies access and displays supported hypothetical conditions and their parameters.

3. The Operator selects one or more conditions, affected Hubs/resources, and a simulation start time and duration.
4. The Operator enters condition-specific values and assumptions, such as arrival rate, demand increase, unavailable capacity, failed points, or vehicle quantity.
5. The system validates required fields, values/units, time period, and resource references.
6. The system displays a summary of conditions, affected resources, assumptions, and period.
7. The Operator confirms; the system saves the configuration as `Ready to Run`, displays its identifier, and confirms no live data changed.

8.2.2 Alternative Flows

A1 — Increased Metro arrivals.
Request the Metro Hub, expected arrival rate/pattern, and affected period.

A2 — Hub at parking capacity.
Request the Hub, simulated occupancy or unavailable-space quantity, and period.

A3 — Charging demand, point failure, or Hub offline.
Request the demand increase or failed points/unavailable Hub and affected period.

A4 — Excessive vehicle concentration.
Request the Hub, vehicle types/quantity, concentration threshold, and period.

A5 — Combined conditions.
Collect each condition's parameters and validate their shared resources and overlapping time periods before displaying the summary.

A6 — Invalid parameters.
Identify the missing, out-of-range, contradictory, or invalid field; the Operator corrects it and repeats validation.

8.2.3 Exception Flows

E1 — Configuration rules unavailable.
Report failure and do not create a ready-to-run scenario.

E2 — Resource unavailable.
Identify a missing/unavailable Hub or resource; do not mark the scenario ready until it is replaced and validated.

E3 — Save failure.
Report that the scenario was not created. Do not start a simulation or change live state.

8.3 UC-W02 — Run a What-if Simulation

Primary actor	Operator
Goal	Run a validated scenario against an isolated, timestamped network snapshot and produce evaluated results and recommendations without changing live operations.
Preconditions	The Operator is authorized; a `Ready to Run` scenario, required network data, and the simulation engine are available.
Trigger	The Operator selects the scenario and requests execution.
Relationships	Includes UC-S06 — Obtain a Network Snapshot, UC-S07 — Evaluate Scenario Impact, and UC-S08 — Generate Coordination Recommendations.
Postconditions	A completed run stores its configuration, baseline, assumptions, results, and recommendations; failed/incomplete output is not published as complete, and live state is unchanged.

8.3.1 Main Flow

1. The system revalidates the scenario and obtains a consistent, timestamped baseline through UC-S06.
2. It creates an identified simulation run and an isolated copy of the relevant state, recording assumptions and data freshness.
3. It runs the scenario asynchronously over the configured period and applies hypothetical events only to simulated state.
4. Through UC-S07, it evaluates service capacity, parking, vehicle, and charging effects relative to the baseline.
5. Through UC-S08, it produces feasible coordination recommendations and expected effects, or records that no intervention is required.
6. It stores the run, results, and recommendations and marks the run `Completed`.
7. The system verifies live state was unchanged and notifies the Operator.

8.3.2 Alternative and Exception Flows

A1 — Live state changes.
Later telemetry does not alter the recorded baseline; freshness and timestamp remain visible.

A2 — Simulated Hub offline.
Active charging becomes `Paused` only in simulated state and simulated charging time stops while offline; live state remains unchanged.

A3 — Limited or no impact.
Record either no material impact/no intervention or affected Hub, period, con-

straint, and remaining unserved demand.

E1 — Snapshot failure.
Do not start with incomplete or mixed baseline data.

E2 — Execution, evaluation, or recommendation failure.
Mark the run `Failed`; do not present partial output as complete.

E3 — Save failure.
Do not publish as completed. All failures leave live operations unchanged.

8.4 UC-W03 — View Simulation Results

Primary actor	Operator
Goal	Inspect predicted service capacity, utilization, vehicle availability, charging congestion, and affected resources with baseline and assumption context.
Preconditions	The Operator is authorized; a completed run and its result/baseline metadata are available.
Trigger	The Operator selects a completed run and chooses View Results.
Postconditions	The completed result is displayed and clearly distinguished from live values; no state is modified.

8.4.1 Main Flow

1. The Operator opens the run list and selects a `Completed` run.
2. The system retrieves the run, scenario, and result data.
3. The system displays run ID, conditions, affected resources, assumptions, baseline timestamp/freshness, and simulation period.
4. The system displays predicted capacity/unserved demand, parking utilization, vehicle availability, charging congestion, and relevant affected resources.
5. The system shows baseline and simulated values using consistent measures, labels predictions as simulated, and visually distinguishes them from live values.
6. The Operator reviews the result.

8.4.2 Alternative and Exception Flows

A1 — No material adverse impact.
Display evaluated measures and indicate that no material adverse impact was predicted.

A2 — Stale baseline data.
Identify affected items and last accepted timestamps; warn that this may affect interpretation while retaining the result.

A3 — Run is not complete.
For a `Running` or `Failed` run, display status/failure reason and do not present partial output as a completed result.

E1 — Run not found.
Report unavailability and do not show another run's values.

E2 — Result or baseline retrieval failure.
Identify unavailable information and do not present incomplete results as complete or current.

8.5 UC-W04 — Review Coordination Recommendations

Primary actor	Operator
Goal	Review proposed redistribution, charging adjustments, or user redirection and their expected effects without applying them to live operations.
Preconditions	The Operator is authorized; a completed run and its recommendation outcome are available.
Trigger	The Operator opens recommendations for a completed simulation run.
Postconditions	Recommendations and effects are displayed; no live assignment, schedule, reservation, Hub state, or routing instruction is changed.

8.5.1 Main Flow

1. The Operator opens a completed run and selects Review Recommendations.
2. The system retrieves the run, impact evaluation, and recommendation outcome.
3. The system displays scenario conditions, affected Hubs/resources, baseline timestamp, and simulation period.
4. The system displays proposed vehicle movements, charging adjustments, or alternative-Hub redirection with expected effects where applicable.
5. The system displays supporting assumptions, addressed constraints, and any remaining unserved demand.
6. The system states that recommendations are simulation outputs and have not been applied live.
7. The Operator reviews the actions and expected effects.

8.5.2 Alternative and Exception Flows

A1 — No intervention required.
Display supporting evaluation measures and the `No Intervention Required` outcome.

A2 — Multiple options.
Display each recommendation for the same constraint separately, with assumptions and expected effects; review does not select or apply an option.

A3 — Live state newer than baseline.
Show the baseline timestamp and warn that recommendations reflect the recorded scenario, not current live state. Do not apply or recalculate automatically.

E1 — Recommendation unavailable.
Report failure; do not infer or show a recommendation from another run.

E2 — Supporting evaluation unavailable.
Mark the affected recommendation incomplete and unsupported; do not apply any live action.

8.6 UC-W05 — Compare What-if Scenarios

Primary actor	Operator
Goal	Compare completed runs, or one run with its baseline, using consistent measures and explicit context.
Preconditions	The Operator is authorized and at least one completed run and its baseline are available; multi-run comparison requires at least two completed runs.
Trigger	The Operator chooses completed runs or a run and its baseline.
Postconditions	A read-only comparison is displayed; scenarios, runs, recommendations, and live state are not modified.

8.6.1 Main Flow

1. The Operator selects completed runs for comparison.
2. The system retrieves each run's conditions, baseline and timestamp, assumptions, period, measures, results, and recommendations.
3. It checks measure definitions/units and identifies differences in snapshots, freshness, assumptions, resources, or time horizons.
4. It displays comparability limitations and valid measures side by side with run and baseline context.
5. It shows differences in service capacity, unserved demand, parking utilization, vehicle availability, charging congestion, and recommendations.
6. The Operator reviews the comparison.

8.6.2 Alternative and Exception Flows

A1 — Baseline comparison.
Compare one completed run with its recorded baseline using the same measures



and units.

A2 — Different context.
Show differing snapshots, horizons, or assumptions; do not claim direct equivalence when not fair.

A3 — No intervention.
Display as a valid outcome with supporting measures, not missing data.

E1 — Run or baseline unavailable.
Identify the missing item and do not substitute data from another run.

E2 — Incompatible measures.
Identify missing/incompatible measures and do not combine them. Produce no comparison if no meaningful common measure remains.

E3 — Retrieval failure.
Do not present an incomplete comparison as complete.

8.7 Simulation Traceability

Requirement	Scenario coverage
FR-NI-09 — Pause Charging When a Hub Is Offline	UC-W02 A2 pauses charging and stops time progression only in the isolated simulation state.
FR-NI-12 — Isolate Simulation From Live Operations	UC-W02 runs against a timestamped snapshot; UC-W03, UC-W04, and UC-W05 are read-only and do not apply recommendations.
NFR — Fail-Safe Simulation States	UC-W02 A2 defines the safe simulated charging state when a Hub is offline.
NFR — Non-Blocking Simulation	UC-W02 runs asynchronously and permits navigation during execution.
NFR — State Transition Logging	UC-W02 records run ID, baseline, state transitions, assumptions, results, and failure stage where applicable.

9 Non-Interactive Functional Requirements

9.1 Purpose and Scope

These requirements specify behaviour initiated by an event, timer, or domain-state change rather than by a Student or Operator selecting a function. They complement the interactive use cases. Thresholds such as battery, data-staleness, reservation-expiry, and Hub-imbalance limits must be configurable rather than embedded in UI components.

9.2 Non-Interactive Functional Requirements

Table 9.1: Non-Interactive Functional Requirements

Requirement	Trigger, required behaviour, and acceptance condition
FR-NI-01 Validate and Apply Vehicle Telemetry *UC-F01, UC-F02, UC-I02*	**Trigger:** Identified telemetry arrives from a provider or mock-IoT timer. **Behaviour:** Validate the vehicle ID, timestamp, and required fields; update location/Hub, battery, operational state, and freshness time only for a valid newer event. **Acceptance:** Invalid or stale telemetry never overwrites newer accepted state.
FR-NI-02 Create or Update a Vehicle-Failure Incident *UC-I02, UC-R03, UC-F04*	**Trigger:** Accepted telemetry reports a fault or ineligible operating state. **Behaviour:** Create one open incident or append evidence to the existing equivalent incident, and restrict the vehicle from new allocation. **Acceptance:** Repeated events do not create duplicate incidents and the vehicle is not offered for reservation or redistribution.
FR-NI-03 Detect Stale Operational Data *UC-F01–UC-F03*	**Trigger:** The latest vehicle, Hub, parking, or charging-point status exceeds the configured staleness interval. **Behaviour:** Mark it stale and expose the last accepted timestamp. **Acceptance:** Operational views distinguish stale information from current information.
FR-NI-04 Expire Unused Vehicle Reservations *UC-R03, UC-R04, UC-R06, UC-R07*	**Trigger:** An unused reservation reaches its configured expiry time. **Behaviour:** Mark it expired and release the allocation when no other commitment or incident prevents availability. **Acceptance:** The reservation cannot be picked up and the vehicle becomes available only when its confirmed condition permits.
FR-NI-05 Synchronize Resource States *UC-R03–UC-R07, UC-P01–UC-P07, UC-F05, UC-I02*	**Trigger:** A confirmed reservation, cancellation, pickup, return, incident, redistribution, parking, or charging event changes availability. **Behaviour:** Recalculate affected vehicle, parking, charging, Hub-occupancy, and assignment states from that event. **Acceptance:** Counts and individual states agree, and no resource is simultaneously available and committed to incompatible work.

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Requirement	Trigger, required behaviour, and acceptance condition
FR-NI-06 Detect Hub Resource Imbalance *UC-F01, UC-F04, UC-W04*	**Trigger:** A confirmed state change occurs or the periodic imbalance check runs. **Behaviour:** Compare availability and utilization with configured thresholds and create an alert or recommendation for shortages, surpluses, or capacity risks. **Acceptance:** The Operator sees the affected Hub and reason; no assignment or physical movement occurs without confirmation.
FR-NI-07 Invalidate an At-Risk Redistribution Assignment *UC-F04, UC-F05, UC-I02*	**Trigger:** An assigned vehicle becomes faulted, unavailable, reserved, rented, or otherwise conflicting. **Behaviour:** Revalidate and flag the assignment and create an operational alert. **Acceptance:** The conflict and reason are visible, and no unconfirmed arrival, location, or occupancy update is recorded.
FR-NI-08 Reconcile Parking Occupancy *UC-H03, UC-P06, UC-P07, UC-F01*	**Trigger:** An accepted parking-occupancy observation arrives. **Behaviour:** Reconcile it with active reservations and sessions, update actual occupancy once per space, and identify discrepancies. **Acceptance:** Occupancy is not double-counted and discrepancies are visible to the Operator.
FR-NI-09 Pause Simulated Charging When a Hub Is Offline *UC-W02, UC-W03*	**Trigger:** A simulated Hub-offline event is applied. **Behaviour:** Pause active simulated charging at that Hub and stop simulated charging-time progression while keeping live state separate. **Acceptance:** No simulated offline-Hub session continues charging and live charging data is unchanged.
FR-NI-10 Guard Scheduled Charging Start *UC-C01–UC-C03*	**Trigger:** A scheduled interval begins or an assigned point changes state. **Behaviour:** Revalidate Hub, point, and allocation availability before starting; otherwise mark the allocation affected or unscheduled with a reason. **Acceptance:** Charging never starts on an unavailable point or offline Hub.
FR-NI-11 Advance the Charging Queue *UC-C01, UC-C02, UC-S04, UC-S05*	**Trigger:** Capacity is released by completion, pause, cancellation, or point availability. **Behaviour:** Reprioritize eligible requests and allocate the next feasible request without overlap; retain reasons for requests that remain unscheduled. **Acceptance:** The next feasible request receives a valid allocation and conflicts are not introduced.

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Requirement	Trigger, required behaviour, and acceptance condition
FR-NI-12 Isolate Simulation From Live Operations *UC-W02–UC-W05*	**Trigger:** A simulation starts, evaluates an event, or finishes. **Behaviour:** Use a consistent timestamped snapshot and apply all changes only to simulation state. **Acceptance:** Results identify their baseline and no live vehicle, reservation, occupancy, charging, or incident state changes because of the simulation.

9.3 Supporting Use-Case Rules

Supporting use case	Rule
UC-S01 — Check Vehicle Availability	Consider operating condition and conflicting commitments, while recognizing the requesting Student's own valid reservation. Return a decision and reason.
UC-S02 — Check Parking Availability	Check occupancy, overlapping reservations, and unavailable spaces in every slot. One unavailable slot makes the whole requested interval unavailable.
UC-S03 — Validate Redistribution Feasibility	Check vehicle eligibility, source/destination suitability, conflicting commitments, and destination capacity.
UC-S04 — Prioritize Charging Requests	Order pending requests by submission timestamp (FCFS); use Request ID as the tie-breaker for identical timestamps. Manual Operator overrides are identified as such.
UC-S05 — Allocate Charging Slots	Assign non-overlapping points and intervals; retain reasons for requests that cannot be scheduled.
UC-S06 — Obtain a Network Snapshot	Capture a consistent timestamped baseline and relevant commitments for simulation.
UC-S07 — Evaluate Scenario Impact	Compare simulated service capacity and resource utilization with the selected baseline.
UC-S08 — Generate Coordination Recommendations	Produce feasible redistribution, charging, or alternative-Hub actions with expected effects, or report that no intervention is required.

9.4 Verification Notes

Requirement group	Representative verification
Telemetry and freshness	Send a valid update followed by an older update; verify that the older event cannot replace current state.
Incident safety	Repeat fault telemetry for one vehicle; verify that one open incident remains and the vehicle receives no new allocation.
Reservation consistency	Let a reservation expire without pickup; verify expiry and a vehicle state consistent with other commitments.
Fleet redistribution	Fault an assigned vehicle before arrival; verify that the assignment is flagged and no completion is recorded.
Charging	Make a point unavailable at scheduled start; verify that charging does not start and the allocation retains a reason.
Simulation isolation	Simulate a Hub-offline event; verify that simulated sessions pause while corresponding live state remains unchanged.

9.5 Traceability

- FR-NI-01–FR-NI-03 support accurate fleet monitoring and incident handling.
- FR-NI-04–FR-NI-05 preserve reservation and resource-state consistency.
- FR-NI-06–FR-NI-07 support redistribution decisions; physical movement still requires Operator confirmation.
- FR-NI-08–FR-NI-11 coordinate parking and charging resources.
- FR-NI-12 keeps What-if Simulation an analysis tool rather than a live-operation command.

10 Non-Functional Requirements

10.1 Functionality

10.1.1 Security (Role-Based Access Control)

The system shall restrict access based on two roles: `Student` and `Operator`. Students cannot access the Operator Dashboard or fleet management tools, and Operators must authenticate to access redistribution and maintenance views.

10.1.2 Input Validation & Error Handling

The system shall validate all reservation and scheduling inputs, preventing bookings for timestamps in the past. Invalid requests must trigger an inline error: *"Invalid time*

selected."

10.1.3 Data Integrity (Concurrency Prevention)

When a vehicle, parking spot, or charging slot is reserved, its status must immediately transition to `Reserved` (locally/in-state) to prevent double-booking from subsequent queries.

10.2 Reliability

10.2.1 Safe Retries (Idempotency)

The system must handle rapid, duplicate user submissions (e.g., double-clicks or unstable connection retries) without creating duplicate booking records.

10.2.2 Session & State Persistence

If a user refreshes the browser during an active flow (e.g., viewing Hub details, selecting a vehicle, or configuring a charging slot), the application must persist the current selection state in client storage (`localStorage` or `sessionStorage`).

10.2.3 Fail-Safe Simulation States

During the What-If Simulation (Domain 6), if an Operator simulates a "Hub Offline" event, all active charging sessions within that Hub must automatically fall back to a `Paused` state rather than counting charging time indefinitely.

10.3 Performance

10.3.1 Client-Side Filtering Latency

Filtering vehicle or hub lists (e.g., filtering by battery level $> 50\%$) must update the view within 1.0 s.

10.3.2 Dashboard Initial Render

The Operator Dashboard overview (hub capacities, active alerts) must achieve an initial render time of under 2.0 s under standard network conditions.

10.3.3 Non-Blocking Simulation

The What-If Simulation engine must compute asynchronously (e.g., via Web Workers or async tasks) without blocking the UI thread, allowing the Operator to navigate other tabs during execution.

10.4 Supportability

10.4.1 Separation of Concerns

The codebase must maintain a clear separation between presentation (HTML/CSS) and domain logic (JavaScript/Python modules) to facilitate parallel work across team members.

10.4.2 State Transition Logging

Key domain state changes (e.g., vehicle changing from `Available` to `Reserved`, charging slot state moving to `Occupied`) must emit structured, timestamped logs to the browser console for MVP debugging.

10.4.3 Cross-Browser Support

The web client must run cleanly on modern evergreen browsers (Chrome, Safari, Edge) without requiring third-party extensions.

10.5 Usability

10.5.1 The 3-Click Rule

Core student user flows (finding a hub and reserving an available vehicle or charging slot) must be completable within a maximum of three navigation steps/clicks from the main view.

10.5.2 Responsive Mobile-First Layout

The student-facing interface must adapt responsively across standard mobile viewport widths (320px to 430px) without horizontal scrolling or layout breakage.

10.6 Constraints (+)

10.6.1 Decoupled Mock Data

All mock/pre-populated datasets (vehicles, hubs, charging stations) must reside in standalone JSON fixtures. Raw data structures must not be hard-coded into UI components.

10.6.2 Timezone & Temporal Standards

All timestamp operations (reservations, charging queues, schedules) must adhere to ISO 8601 formatting and be normalized to the local timezone (`GMT+7`) to prevent queue calculation errors.

10.6.3 Lightweight Smart Charging Logic

The charging scheduler uses straightforward rule-based prioritization (e.g., lowest battery first or first-come, first-served). Each hub manages a queue of up to 20 vehicles at a time, ensuring slot assignments update smoothly in the browser without requiring complex optimization engines.

10.6.4 Mock IoT

Vehicle telemetry (GPS coordinates, battery levels, digital locks) shall be simulated via state events and mock timers; no live hardware integration is required.

10.6.5 Simulated Billing

Rental and charging transactions shall operate exclusively via a mock student campus credit system without external third-party payment gateways.

References

[1] Object Management Group. *OMG Unified Modeling Language, Version 2.5.1*. Object Management Group, 2017. Section 18: Use Cases. Available at: https://www.omg.org/spec/UML/2.5.1.

[2] Roger S. Pressman and Bruce R. Maxim. *Software Engineering: A Practitioner's Approach*. McGraw-Hill Education, New York, NY, 9th edition, 2020.

[3] Ian Sommerville. *Software Engineering*. Pearson Education, Boston, MA, 10th edition, 2016.